

Foundations & Basics

Introduction to SQL & SQL Command Types



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What is SQL & Why it Matters

- SQL: **S**tructured **Q**uery **L**anguage
- Standard language for relational databases
- Allows creating, reading, updating, deleting data
- Developed in the 1970s at IBM
- Became the foundation for modern relational databases



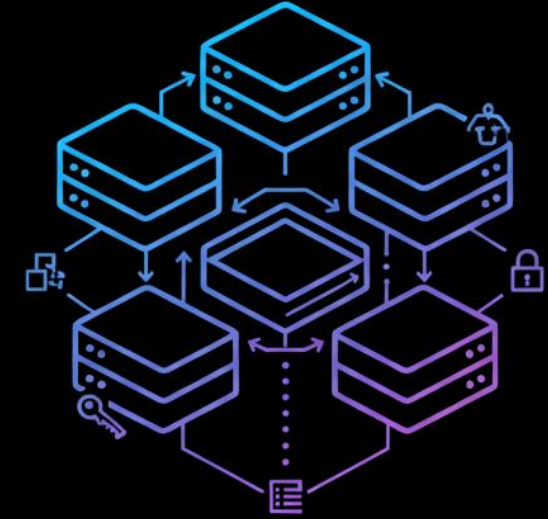
What is a DBMS & How It Differs from an RDBMS

- DBMS: Software to manage databases; may not enforce relationships
 - Example: NoSQL Databases
- RDBMS: Relational DBMS, stores data in tables with relationships

Examples: SQL Server, MySQL, PostgreSQL

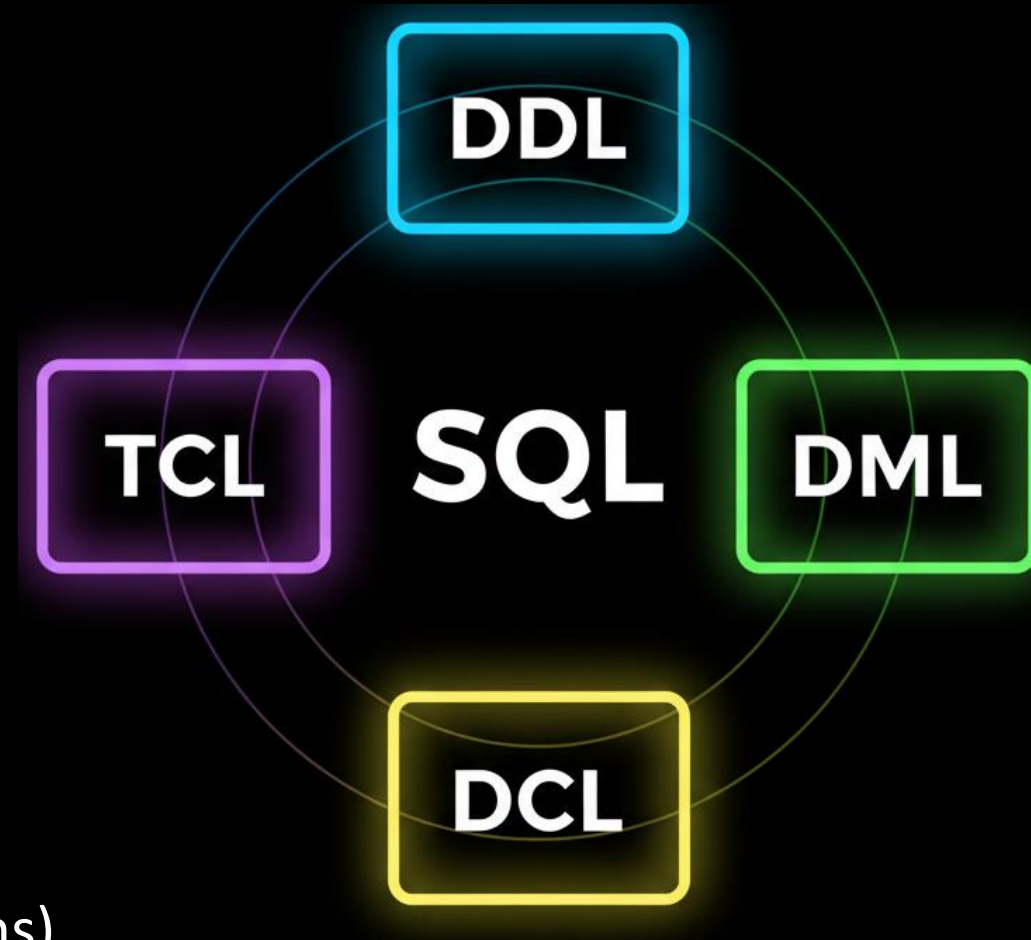


- Every RDBMS is a DBMS, but not every DBMS is an RDBMS
- DBMS Handles storage, retrieval, and security of data



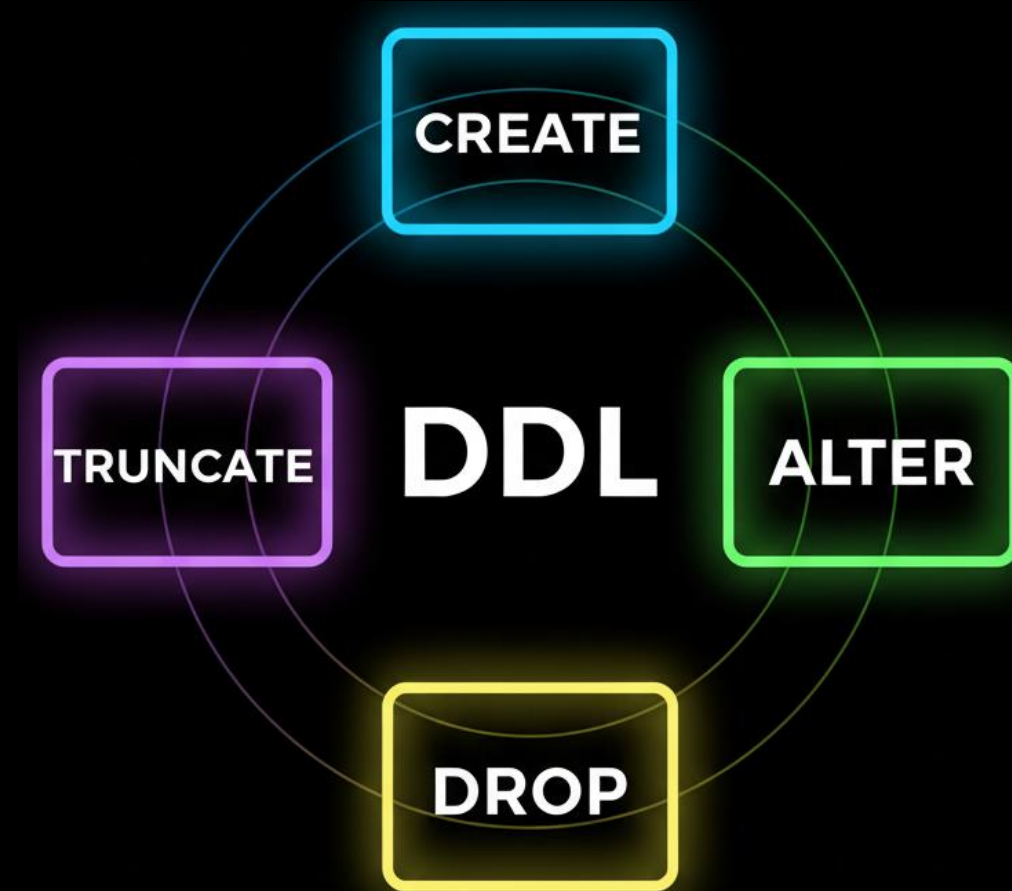
SQL Command Categories

- **DDL**
Data Definition Language (structure of tables)
- **DML**
Data Manipulation Language (working with data)
- **DCL**
Data Control Language (permissions)
- **TCL**
Transaction Control Language (managing transactions)



Data Definition Language (DDL)

- **CREATE**
Build database objects (tables, views, indexes)
- **ALTER**
Modify database objects
- **DROP**
Delete database objects
- **TRUNCATE**
Clear all data from a table



Data Manipulation Language (DML)

- **SELECT**
Retrieve data
- **INSERT**
Add records
- **UPDATE**
Modify records
- **DELETE**
Remove records



Data Control Language (DCL)

- **GRANT**
Assign permissions
- **REVOKE**
Remove permissions



Transaction Control Language (TCL)

- **COMMIT**
Save changes
- **ROLLBACK**
Undo changes
- **SAVEPOINT**
Set checkpoints

